



NSF GRFP: Tips & Tricks

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Overview

- My Story
- General advice
- Tools
- Statistics
- Important considerations
- Research proposal
- Immediate Advice



My Story

- BS in Mechanical Engineering
- Internships, clubs, school research
- I started working on my application in the middle of the summer (I did not know about this previously)
 - I brainstormed ideas for about a month and started working on one
 - About a month later, I decided to change 70% of my research direction
 - Moral of the story: do not fall for the sunk cost fallacy
- I submitted my application about 10 minutes before it was due
 - There were VERY minor changes over the previous two days



General Advice — Content

- Do it! It is a fantastic opportunity
- Your application is just your transcript, your 3-page personal statement and your 2-page research proposal
 - No CV
 - No resume
 - No cover letter
- You need to communicate everything that the reviewers will need to know about you in those 5 pages



General Advice — Reviewers

- Find any professors at your university that have reviewed NSF GRFP proposals
 - For mechanical engineers: Mark Colton used to review
 - If you can't find any, then find professors who have had students received the NSF GRFP. Or any research-focused professor will work too!
 - Have your professors read your application and iterate multiple times
- Have your friends, family, colleagues read it
 - Your reviewer is guaranteed to be in the same major field but not necessarily the same subfield (I could have an electrical engineer read my robotics proposal)
- Follow the solicitation guidelines and rules



General Advice — Personal Background

- If you do not have a tragic story, you do not need one
 - My application did not focus much on hardships, as I do not feel like I have too many hardships that set me apart (college is tough, etc)
 - Instead, I focused on what I have tried to accomplish with my time, being proactive about my education and involvement in STEM
- Realize what unique experiences you have as a BYU student
 - Many other universities do not have undergraduate research opportunities
 - Missions/church service
 - Study abroads
 - TAs



General Advice — IM and BI

- Do anything you can this summer/early Fall to expand your Intellectual Merit and Broader Impacts
 - Volunteer, publish papers, research your topic, make educational videos, do outreach to schools
- Your broader impacts and intellectual merit need to be measurable
 - When the NSF asks for their yearly update, how are you going to prove that you have met your goals? YouTube views, how you are distributing open-source projects, number of students mentored, etc
 - The reviewers will not like a goal that is either 1) too easy or 2) impossible



General Advice — Formatting

- Be creative with bolding and italics
 - Inline italicized headings save space while still separating ideas/sections
- Include citations
 - Back up your idea with the scientific findings of others
 - “[1] Wilk-Jakubowski et al. Technology in Society, 2022” is all you need (name, journal, year)



Tools

- Examples from previous years: <https://www.alexhunterlang.com/nsf-fellowship>
 - This website only has examples from 2022 and earlier. The NSF format and rules have changed a little, but still good reference material
- A newer website with fewer, but more recent, examples: <https://grfp-help.github.io/>
- NSF GRFP Reddit: both applicants and reviewers check this and can answer questions
- Solicitation: <https://www.nsf.gov/funding/opportunities/grfp-nsf-graduate-research-fellowship-program/nsf25-547/solicitation>
 - There are examples of ineligible research topics



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133 \newcommand{\comment}[1]{%
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143 % -----
144 % BEGINNING OF THE DOCUMENT
145 % -----
146 \begin{document}
147
148 \begin{center}
149 \large{\bf Communication-Adaptive CBFs for Multi-Agent
150 Navigation -- Research Statement}
151 \end{center}
152
153 \section*{Motivation and Problem Statement:}
154 Teams of robots are increasingly deployed in disaster
155 response scenarios --- collapsed buildings, subterranean
156 sites, and wildfire zones --- with rapidly changing,
157 communication-limited conditions
158 \cite{WILKJAKUBOWSKI2022101935}. Communication failures
159 degrade coordination, causing redundant coverage,
160 incomplete mapping, and unsafe behaviors. Because lives may

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**Motivation and Problem Statement:** Teams of robots are increasingly deployed in disaster response scenarios — collapsed buildings, subterranean sites, and wildfire zones — with rapidly changing, communication-limited conditions [1]. Communication failures degrade coordination, causing redundant coverage, incomplete mapping, and unsafe behaviors. Because lives may depend on rapid, accurate search, **ensuring reliable coordination under imperfect communication is mission-critical for saving lives**. Existing multi-robot methods handle communication loss in various ways, from reactive recovery [2] to conservative connectivity maintenance [3], yet few explicitly incorporate *predicted* communication quality into safety-critical control. This leaves a critical gap: *maintaining safety constraints (collision avoidance, connectivity) under unreliable communication*. My research addresses this challenge with **Communication-Adaptive Control Barrier Functions (CA-CBFs)**, which predict communication quality and adjust safety constraints and paths in real time. The framework enables multi-agent teams to explore efficiently, coordinate safely despite intermittent or noisy communication, and adapt to unknown environments. While motivated by disaster response, CA-CBFs can also apply to remote terrestrial and space exploration.

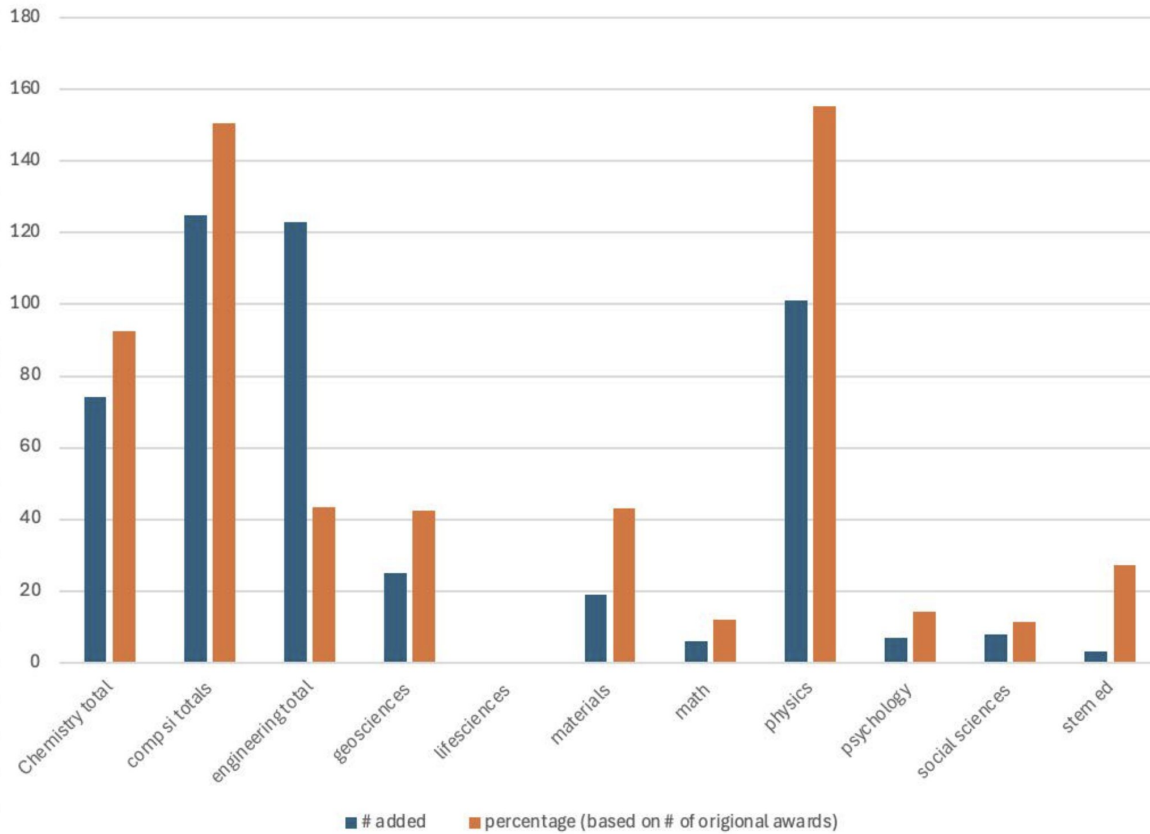
**Proposed Solution & Research Plan:** My research will develop Communication-Adaptive Control Barrier Functions (CA-CBFs), a novel framework where multi-agent safety constraints adjust to predicted communication





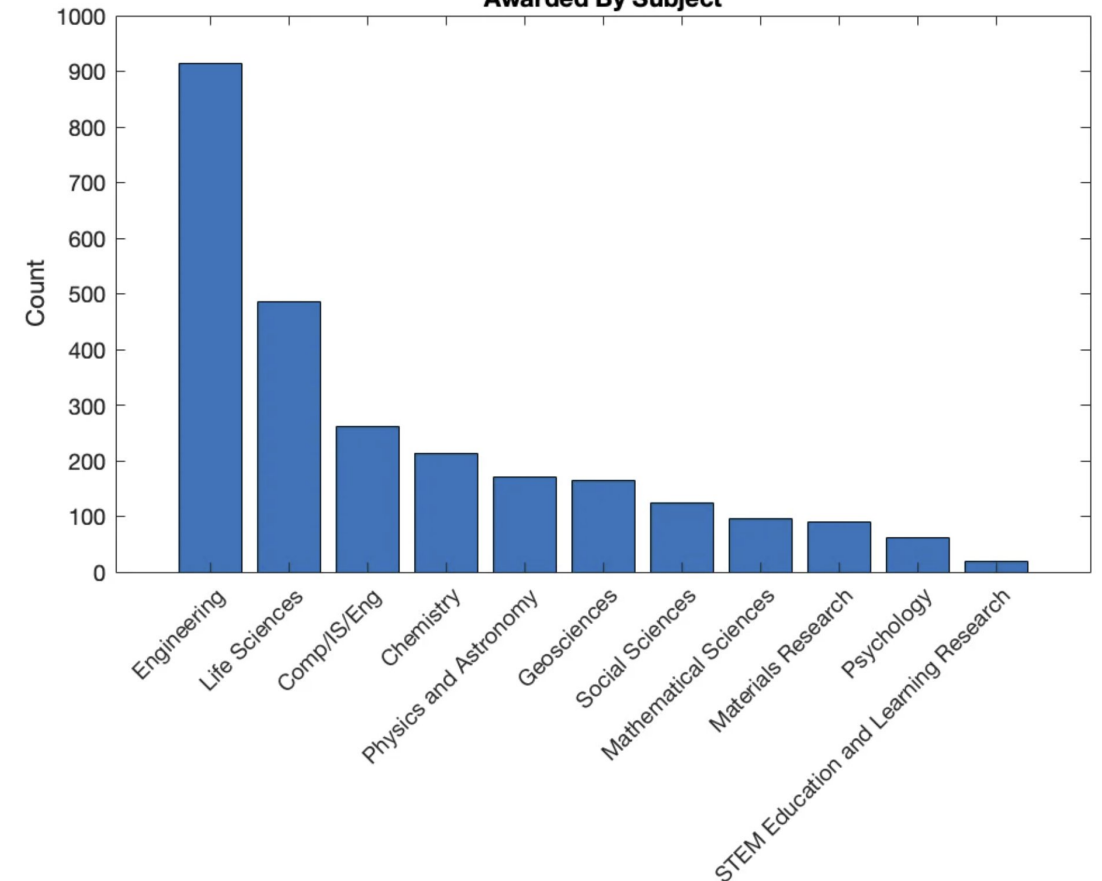
## 2024-2025 cycle

GRFP new awards by field



## 2025-2026 cycle

Awarded By Subject





# Important Considerations

- Your graduate degree cannot be
  - A professional degree (Masters in Engineering vs Masters in Science)
  - Primarily course-based with no thesis requirement

Only eligible research-based master's and doctoral degrees in STEM, including STEM education, are eligible for NSF GRFP support. Professional degree programs and graduate programs that are primarily course-based with no thesis requirement are ineligible for NSF GRFP support. **No clinical or health degree programs are eligible for NSF GRFP consideration. In FY2026, Clinical Psychology graduate degree programs are not eligible for NSF GRFP.**

<https://www.nsf.gov/funding/opportunities/grfp-nsf-graduate-research-fellowship-program/nsf25-547/solicitation>



# Important Considerations

- With the current administration, there are some things to avoid:
  - Do not mention national protected classes (gender, race, ethnicity, age, etc)
    - My research cannot study the effects of X on the African-American community
    - I cannot propose an engineering idea to assist the elderly community
  - Outside of these classes, you can mention specifics
    - Socio-economic status, education status, etc
    - However, be careful not to combine non-protected classes to suspiciously look like a protected class
  - Avoid politically charged words
    - An example given to me by the NSF was replacing “climate change” or “global warming” to “changes in the weather”
- Gossip: include something about AI (supposedly, there is a recent-ish executive order for the NSF to weight proposals with AI higher)



# Research Proposal — Layout

- My proposal's layout:
  - Motivation and Problem Statement
    - Background & Research Gap
  - Proposed Solution & Research Plan —> Give a rough timeline with structured goals
    - Objective 1
    - Objective 2
    - Objective 3
    - Objective 4
  - Intellectual Merit
  - Broader Impacts —> Use the language for broader impacts on the NSF website
    - Societal Impacts and Partnerships
    - Open-Source Materials and STEM Education
    - Mentoring and Outreach
  - References —> Citations! I had 10





# Research Proposal — Research Idea

- You do not need to do this idea! I probably will not do mine
- The idea generally needs to balance being applied and research-focused. This often depends on how you frame your idea.
  - My proposal cannot be pitched as building a new drone — there is no obvious research questions here
  - You could instead pitch an idea to solve the issue of drones getting lost in wildfires due to GPS interference
    - Relevance: wildfires and disaster response
    - Theoretical question: autonomous navigation, multiple robot control
    - Application: demonstrable on hardware or in simulation, and may involve building custom hardware
- Provide a timeline. A 3-year project at minimum



# Research Proposal — Figures

- Make a figure that explains your idea, and make it pop!

## Communication-Adaptive CBFs for Multi-Agent Navigation – Research Statement

**Motivation and Problem Statement:** Teams of robots are increasingly deployed in disaster response scenarios — collapsed buildings, subterranean sites, and wildfire zones — with rapidly changing, communication-limited conditions [1]. Communication failures degrade coordination, causing redundant coverage, incomplete mapping, and unsafe behaviors. Because lives may depend on rapid, accurate search, **ensuring reliable coordination under imperfect communication is mission-critical for saving lives**. Existing multi-robot methods handle communication loss in various ways, from reactive recovery [2] to conservative connectivity maintenance [3], yet few explicitly incorporate *predicted* communication quality into safety-critical control. This leaves a critical gap: *maintaining safety constraints (collision avoidance, connectivity) under unreliable communication*. My research addresses this challenge with **Communication-Adaptive Control Barrier Functions (CA-CBFs)**, which predict communication quality and adjust safety constraints and paths in real time. The framework enables multi-agent teams to explore efficiently, coordinate safely despite intermittent or noisy communication, and adapt to unknown environments. While motivated by disaster response, CA-CBFs can also apply to remote terrestrial and space exploration.

**Background & Research Gap:** Current multi-robot methods typically treat safety and communication-based path planning separately. Control Barrier Functions (CBFs) ensure robots stay within safe boundaries by defining a *barrier function*  $h(x)$  with  $h(x) \geq 0$  indicating safe states. Many CBF-based multi-robot methods assume reliable communication or enforce fixed connectivity [3, 4], yet real environments feature intermittent signals that affect coordination capabilities. While some CBF variants adapt to time-varying conditions [5, 6], they do not adapt safety constraints based on *predicted* communication degradation. Additionally, communication-aware planners predict signal quality [7, 8] but lack real-time safety guarantees. **CA-CBFs bridge this gap by embedding predictive communication models directly into the barrier function framework.** Each robot can proactively adjust its behavior and coordinate with teammates, maintaining safe operation even when connectivity is uncertain or asymmetric. This approach unifies predictive communication modeling and real-time, safety-critical control for multi-agent systems.

**Proposed Solution & Research Plan:** My research will develop Communication-Adaptive Control Barrier Functions (CA-CBFs), a novel framework where multi-agent safety constraints adjust to predicted communication quality  $c(t) \in [0, 1]$  — representing signal strength, bandwidth, and connectivity.

The key innovation is the barrier function  $h(x, c) = d_{\max}(c)^2 - d(\text{robot}_i, \text{robot}_j)^2$ , where the maximum allowable separation  $d_{\max}(c)$  between robots  $i$  and  $j$  grows when communication is strong ( $c(t) \approx 1$ , enabling aggressive exploration) and shrinks as it degrades ( $c(t) \approx 0$ , for conservative coordination), as shown in Figure 1. Each agent predicts

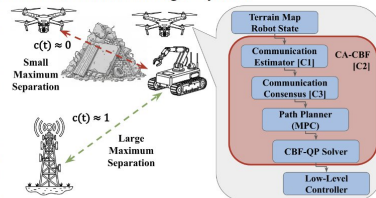


Figure 1: CA-CBF concept for disaster response. Communication quality  $c(t)$  adapts separation limits. Predicted  $c(t)$  informs planning, and CA-CBFs minimally adjust controls for safety. The three core challenges (C1, C2, C3) are labeled in context.

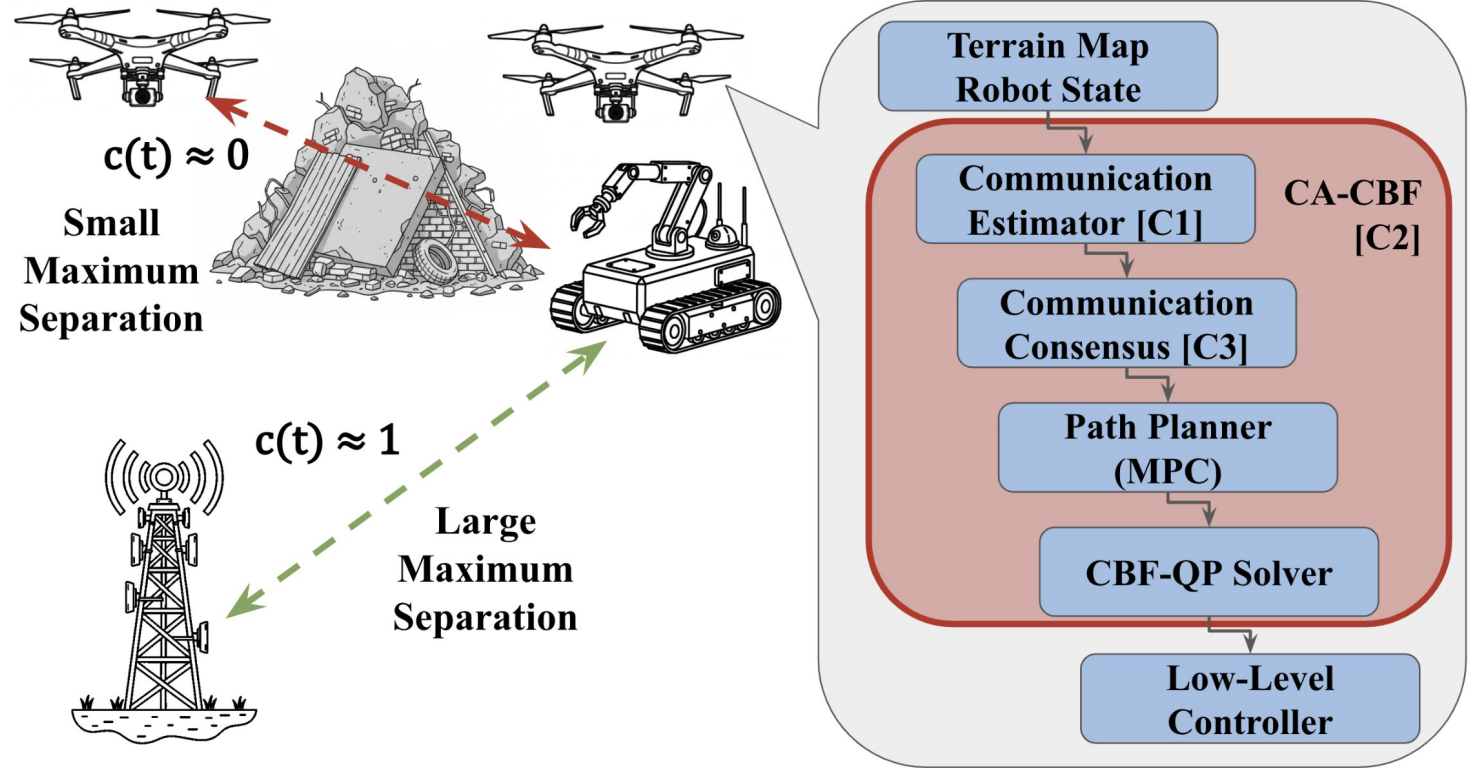


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# Research Proposal — IM and BI

- For your personal statement, your intellectual merit and broader impacts should primarily focus on what you have already demonstrated, with mentions of future, overarching goals as a researcher
  - The intellectual rigor you have demonstrated
  - The broader impacts you have strived to achieved
  - Your goals as a future professor/researcher/etc to continue these impacts
- For your research proposal, your intellectual merit and broader impacts should primarily focus on your research project
  - The intellectual advances of your research
  - The broader, societal impacts of your research



# Research Proposal — IM and BI

- Intellectual Merit:
  - Demonstrate that your proposal has intellectual merit by 1) showing the research gap and 2) showing how you will fill that gap
- Broader Impacts:
  - Use the exact wording found here: <https://www.nsf.gov/funding/learn/broader-impacts> for examples of broader impacts



# My Immediate Advice

- What can you do this summer to “boost” your broader impacts and intellectual merit?
- Start building relationships with reference letter writers (you need two, three is recommended)
- Start brainstorming your topic
  - Talk with AI models to generate ideas
  - Find 3-5 ideas you like, flesh out the rough idea of each one, and send them to professors in your field for their feedback
  - The research proposal took me longer to formulate, write, and trim down



# Any Questions?

Thank you